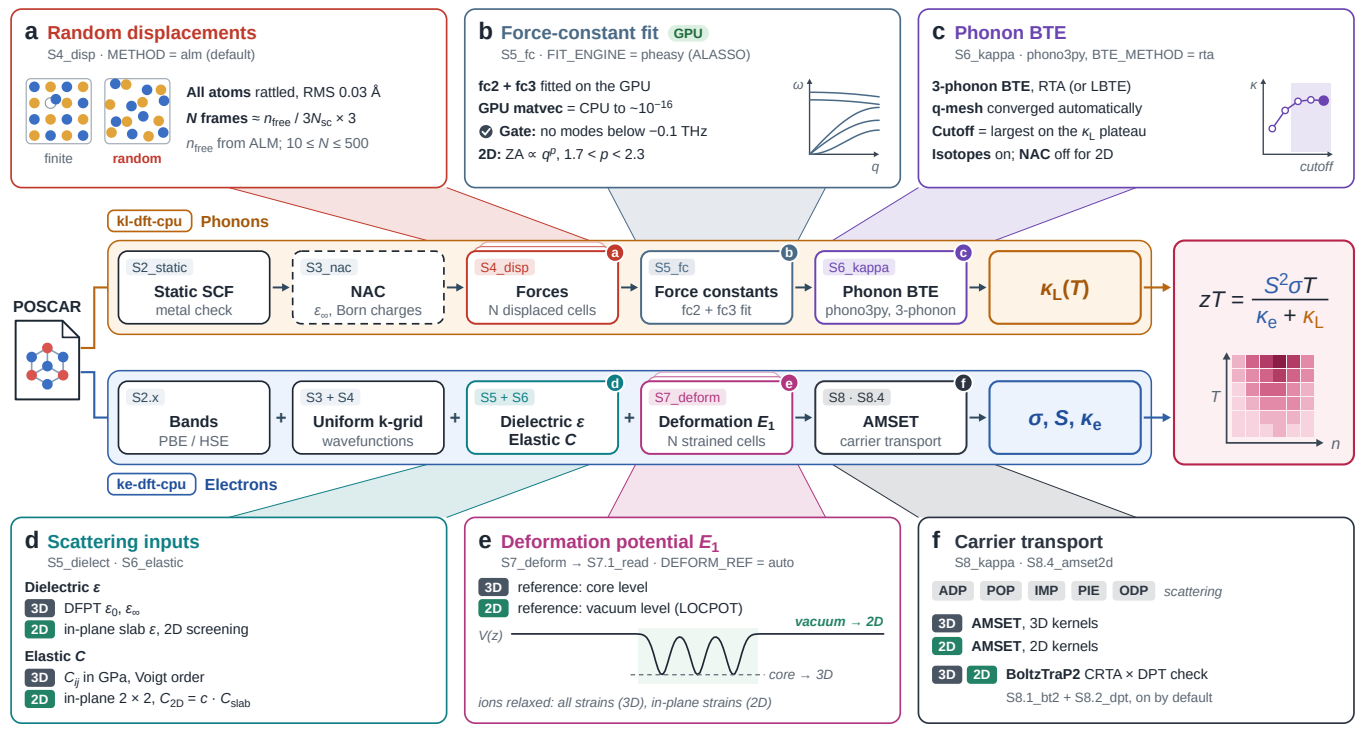

a Random displacements
S4_disp · METHOD = alm (default)
finite
random
All atoms rattled, RMS 0.03 Å
N frames ≈ n_free / 3N_sc × 3
n_free from ALM; 10 ≤ N ≤ 500
b Force-constant fit GPU
S5_fc · FIT_ENGINE = pheasy (ALASSO)
fc2 + fc3 fitted on the GPU
GPU matvec = CPU to ~10⁻¹⁶
Gate: no modes below −0.1 THz
2D: ZA ∝ q^p, 1.7 < p < 2.3
ω
q
c Phonon BTE
S6_kappa · phono3py, BTE_METHOD = rta
3-phonon BTE, RTA (or LBTE)
q-mesh converged automatically
Cutoff = largest on the κ_L plateau
Isotopes on; NAC off for 2D
κ
cutoff
kl-dft-cpu Phonons
POSCAR
S2_static
Static SCF
metal check
S3_nac
NAC
ε∞, Born charges
S4_disp
Forces
N displaced cells
a
S5_fc
Force constants
fc2 + fc3 fit
b
S6_kappa
Phonon BTE
phono3py, 3-phonon
c
κL(T)
S2.x
Bands
PBE / HSE
S3 + S4
Uniform k-grid
wavefunctions
S5 + S6
Dielectric ε
Elastic C
d
S7_deform
Deformation E1
N strained cells
e
S8 · S8.4
AMSET
carrier transport
f
σ, S, κe
ke-dft-cpu Electrons
zT = S²σT / (κe + κL)
T
n
d Scattering inputs
S5_dielect · S6_elastic
Dielectric ε
3D DFPT ε0, ε∞
2D in-plane slab ε, 2D screening
Elastic C
3D Cij in GPa, Voigt order
2D in-plane 2 × 2, C2D = c · Cslab
e Deformation potential E1
S7_deform → S7.1_read · DEFORM_REF = auto
3D reference: core level
2D reference: vacuum level (LOCPOT)
V(z)
vacuum → 2D
core → 3D
ions relaxed: all strains (3D), in-plane strains (2D)
f Carrier transport
S8_kappa · S8.4_amset2d
ADP POP IMP PIE ODP scattering
3D AMSET, 3D kernels
2D AMSET, 2D kernels
3D 2D BoltzTraP2 CRTA × DPT check
S8.1_bt2 + S8.2_dpt, on by default